

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)	Use of period equation (1) Algebra (minimum $d^3 = \frac{GMT^2}{4\pi^2}$) (1) Answer: $d = 1.57 \times 10^{11}$ m (1) ecf on T N.B. deriving the equation is ok Don't accept $v^2 = \frac{GM}{r}$ with tiny speed	1	1 1		3	2	
		(ii)	Use of C of M equation N.B. be generous on substitution (1) Mass = 3×10^{25} [kg] (1) ecf	1	1		2	2	
	(e)		1× bad point / something to do with mass e.g. gravity stronger, no idea of atmosphere etc. don't accept gravity between star and planet greater 1× good point / linked to intensity of light e.g. temperature about right, habitable zone, water possible 1× conclusion linked to a point (even if the point isn't quite deserving of the mark e.g. No, it's too hot - 1 mark) e.g. Yes because water possible (this is 2 marks) Accept 2 different conclusions or even sitting on the fence			3	3		
			Question 3 total	4	9	3	16	9	0